

Global Options

`--json` JSON output to stdout
`-v, --verbose` Show operational details
`-l, --log` Per-file progress to stderr
`-d, --debug` External tool debug output
`-t, --time` Show elapsed time

Setup

maki init
Create a new catalog in the current directory.

maki volume add [LABEL] PATH
Register a storage volume (label auto-derived from path if omitted).
`--purpose` media|working|archive|backup|cloud

maki volume list
Show registered volumes with online/offline status.
`--purpose` PURPOSE `--offline` `--online`

maki volume set-purpose / rename / remove
Change purpose, rename, or unregister a volume.

maki volume combine / split
Merge two volumes or split a subdirectory into new volume.

Ingest & Edit

maki import PATHS...
Import files into the catalog.
`--volume` `--profile` NAME `--auto-group` `--smart`
`--embed` [Pro] `--describe` [Pro] `--dry-run`
`--include` GROUP `--skip` GROUP

maki delete ASSET_IDS...
Delete assets. `--apply` `--remove-files`

maki tag ASSET_ID TAGS...
Add or remove tags. `--remove`

maki tag rename OLD NEW
Rename a tag across all assets. `--apply`
OLD accepts = (exact level) and ^ (case-sensitive) prefixes.

maki tag delete TAG
Delete a tag (cascades to descendants) from every asset. `--apply`
TAG accepts the same = / ^ markers as **tag rename**.

maki tag clear ASSET_ID
Remove all tags from an asset.

maki tag fix-unicode
NFC-normalise every tag value catalog-wide; merges NFC/NFD duplicates per asset. `--apply`

maki tag export-vocabulary
Export tag tree as vocabulary.yaml.
`--prune` `--default` `--output` FILE `--counts`
`--format` yaml|text|json (text = tab-indented for Lightroom / Capture One; json = nested objects for scripts; `--counts` adds ‘# N assets’ trailing YAML comments)

maki edit ASSET_ID
Edit metadata fields.
`--name` `--description` `--rating` N
`--label` COLOR `--date` `--clear`-FIELD

maki group HASHES...
Group variants into one asset.

maki split ASSET_ID --variant HASH
Split variants out of an asset.

maki auto-group [QUERY]
Auto-group by filename stem (directory-local). `--apply` `--global`

maki describe [QUERY] [Pro]
Generate VLM image descriptions.
`--asset` ID `--volume` `--model`
`--mode` describe|tags|both
`--apply` `--force` `--dry-run`

maki auto-tag [QUERY] [Pro]
AI auto-tagging with SigLIP.
`--asset` `--volume` `--threshold`
`--labels` FILE `--apply` `--download`

maki embed [QUERY] [Pro]
Generate image embeddings.
`--asset` `--volume` `--force` `--export`

Organize

maki collection create / add / remove / list / show / delete
Static albums. **create** NAME, **add** NAME IDS..., **remove** NAME IDS...

maki saved-search save / list / run / delete
Dynamic smart albums. **save** NAME QUERY `--sort` `--favorite`

maki stack create ASSET_IDS...
Create a stack. First = pick.

maki stack add REF ASSET_IDS...
Add assets to existing stack.

maki stack remove / pick / dissolve / list / show
Stack management commands.

maki stack from-tag PATTERN
Convert tag patterns to stacks. `--apply` `--remove-tags`

maki faces detect / cluster / clean / name / merge [Pro]
Face detection and recognition.
`--query` `--asset` `--apply` `--force` `--threshold` `--min-confidence`
Diagnostics: **faces similarity**, **faces dump-aligned**, **faces status**

Retrieve

maki search QUERY
Search assets with filters (see back).
`--format` ids|short|full|json|{template}
`-q` (shorthand for `--format ids`)

maki show ASSET_ID
Show full asset details.

maki preview ASSET_ID
Open preview in OS default image viewer.

maki export QUERY TARGET
Export to directory. `--zip` `--layout` flat|mirror
`--all-variants` `--sidecars`

maki contact-sheet QUERY OUTPUT
Generate PDF contact sheet.
`--layout` `--paper` `--landscape`
`--title` `--fields` `--sort`

maki duplicates
Find duplicate files. `--mode` same|cross `--volume` `--format`

maki stats
Catalog statistics breakdown. `--tags` `--verified`

maki status
Health overview with next-step hints. `--min-copies` N

maki backup-status
Show backup coverage and at-risk assets. `--copies` N

maki doc [manual/cheatsheet/filters]
Open documentation PDF in browser.

maki licenses
Show MAKI license + 3rd-party crate licenses. `--summary`

maki serve
Start web UI. `--port` `--bind`

maki shell
Interactive shell. `-c` CMD `--script` FILE

Maintain

maki verify [PATHS...]
Check file integrity.
`--volume` `--asset` `--force` `--max-age` DAYS

maki sync [PATHS...]
Sync with disk changes. `--apply` `--remove-stale`

maki sync-metadata [QUERY] [Pro]
Bidirectional XMP sync. `--asset` `--dry-run`

maki refresh [PATHS...]
Re-read changed recipes. `--dry-run` `--media`
`--reimport` — full re-extract `--exif-only` — EXIF only

maki writeback [QUERY] [Pro]
Write pending edits to XMP. `--all` `--mirror-tags` `--force` `--dry-run`
`--volume` `--asset`
`--mirror-tags` drops XMP entries the catalog doesn’t have (requires `--all`).
`--force` rebuilds dc:subject + lr:hierarchicalSubject from catalog (more aggressive).

maki cleanup
Remove stale records. `--apply` `--list` `--path`

maki dedup

Remove same-volume duplicate file locations. --apply --prefer

maki generate-previews [PATHS...]

Generate/regenerate previews.

--force --smart --upgrade

--volume --asset --include --skip

maki relocate [ASSET_IDS...]

Copy/move files between volumes.

--target VOLUME --query --remove-source

--create-sidecars --dry-run

maki rebuild-catalog

Rebuild SQLite from YAML sidecars. --asset ID (per-asset)

Also: migrate, create-sidecars, update-location, fix-roles,
fix-dates, fix-recipes

Core Concepts

Asset	Logical media item (one photo)
Variant	Concrete file (Original, Alternate, Processed, Export)
Recipe	XMP/processing sidecar (tracked by location)
Volume	Storage device (online/offline detection)
Stack	Collapsed group in browse (own metadata)
Collection	Static album (manual curation)
Saved Search	Dynamic smart album (query re-evaluated)

Key Workflows

- Import & Organize
1. `maki init` + `maki volume add`
 2. `maki import --smart /path`
 3. `maki serve` — browse in web UI
 4. Rate, tag, label via UI or CLI
 5. `maki auto-tag ""` [\[Pro\]](#)— AI tagging

- Maintenance Cycle
1. `maki verify` — check integrity
 2. `maki sync` — detect moved/missing
 3. `maki sync-metadata` — bidirectional XMP [\[Pro\]](#)
 4. `maki cleanup --apply` — remove stale

- Backup & Archive
1. `maki volume add Archive /mnt/nas --purpose backup`
 2. `maki relocate --target Archive ""` — copy all
 3. `maki backup-status` — check coverage
 4. `maki dedup --apply` — remove same-volume dups

- Export & Deliver
1. `maki export "rating:4+ tag:portfolio" ./out`
 2. `maki export "collection:Client" ./out --zip`
 3. `maki contact-sheet "rating:5" portfolio.pdf`

Search Filter Syntax

All filters work in CLI `maki search` and web UI search box.

Text & Metadata	
<code>tag:landscape</code>	Tag match (any level)
<code>tag:= / tag:/</code>	= whole path (exact value); /leaf only
<code>tag:~Landscape</code>	Case-sensitive (not “landscape”)
<code>tag: wed</code>	Prefix anchor (wedding, wedding-2024, ...)
<code>-tag:rejected</code>	Exclude tag
<code>type:image</code>	Asset type
<code>format:nef</code>	File format
<code>label:Red</code>	Color label
<code>camera:fuji</code>	Camera (partial)
<code>lens:56mm</code>	Lens (partial)
<code>description:cat</code>	Description (substring; alias <code>desc:</code>)
<code>person:Alice</code>	Named person (repeat for AND, comma for OR) [Pro]
<code>collection:Fav</code>	Collection
<code>path:2026/</code>	Path prefix (* = wildcard)
<code>id:72a0</code>	Asset ID prefix
<code>meta:key=val</code>	Source metadata

Numeric Filters

All support: `x` (exact), `x+` (min), `x-y` (range), `x,y` (OR)

<code>rating:3+</code>	3 stars or more
<code>rating:3-5</code>	3 to 5 stars
<code>rating:2,4</code>	Exactly 2 or 4
<code>rating:0</code>	Unrated (“the rest”)
<code>iso:100-800</code>	ISO range
<code>focal:35-70</code>	Focal length range
<code>f:1.4-2.8</code>	Aperture range
<code>width:4000+</code>	Min width (px)
<code>height:2000+</code>	Min height (px)
<code>copies:2+</code>	File copies
<code>variants:2+</code>	Variant count
<code>scattered:2+</code>	Files spread across 2+ session roots
<code>duration:30+</code>	Video duration (seconds)
<code>codec:h264</code>	Video codec (partial)
<code>faces:2+</code>	Face count [Pro]

Date Filters

<code>date:2026</code>	Year prefix
<code>date:2026-03</code>	Month prefix
<code>date:2026-03-15</code>	Exact date
<code>dateFrom:2026-01</code>	From (inclusive)
<code>dateUntil:2026-12</code>	Until (inclusive)

Status Filters

<code>orphan:true</code>	No file locations
<code>orphan:false</code>	Has file locations
<code>missing:true</code>	Files missing on disk
<code>stale:30</code>	Not verified in N days
<code>stacked:true</code>	In a stack
<code>volume:Archive</code>	On specific volume
<code>volume:none</code>	No online volume
<code>geo:any</code>	Has GPS data

Pro Filters

<code>similar:72a0bb4b</code>	Visual similarity
<code>min_sim:90</code>	Similarity threshold
<code>text:sunset</code>	Text-to-image search
<code>embed:none</code>	No embedding

Combining Filters

<code>tag:a,b</code>	a OR b (comma)
<code>tag:a tag:b</code>	a AND b (repeat)
<code>-tag:x</code>	NOT (dash prefix)

Output Formats

<code>maki search --format PRESET/TEMPLATE</code>	
<code>ids</code>	One UUID per line
<code>short</code>	ID, name, type, format, date
<code>full</code>	+ tags, description, rating
<code>json</code>	JSON array
<code>'{id}\t{name}'</code>	Custom template
Placeholders: {id}, {name}, {filename}, {type}, {format}, {tags}, {description}, {rating}, {label}, {date}, {size}	

Configuration (maki.toml)

<code>[preview]</code>	<code>max_edge</code> , <code>format</code> , <code>quality</code> , <code>smart__max_edge</code>
<code>[serve]</code>	<code>port</code> , <code>bind</code> , <code>per_page</code>
<code>[import]</code>	<code>exclude</code> , <code>auto_tags</code> , <code>smart__previews</code> , <code>profiles</code>
<code>[browse]</code>	<code>default_filter</code>
<code>[cli]</code>	<code>log</code> , <code>time</code> , <code>verbose</code> (default global flags)
<code>[writeback]</code>	<code>enabled</code> (default: false) [Pro]
<code>[group]</code>	<code>session_root_pattern</code> (auto-group scope)
<code>[dedup]</code>	<code>prefer</code>
<code>[verify]</code>	<code>max_age_days</code>
<code>[ai]</code>	<code>model</code> , <code>threshold</code> , <code>labels</code> [Pro]
<code>[vlm]</code>	<code>endpoint</code> , <code>model</code> , <code>timeout</code> [Pro]
<code>[contact_sheet]</code>	<code>layout</code> , <code>paper</code> , <code>fields</code>

